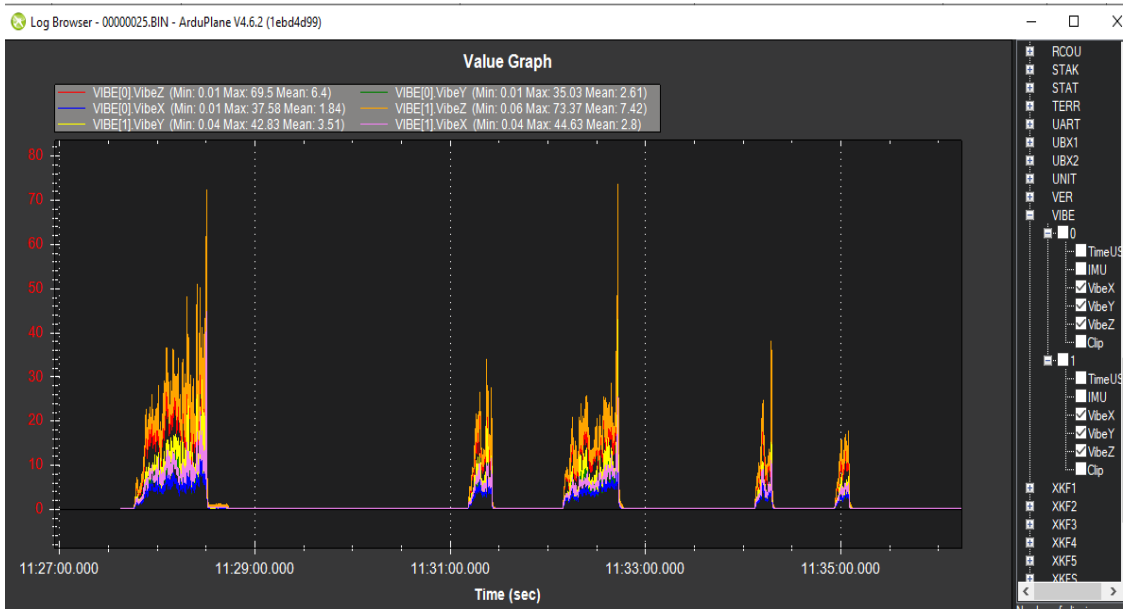
# Measuring Vibration from Log

Autopilots have accelerometers that are sensitive to vibrations. These accelerometer values are combined with barometer and GPS data to estimate the vehicle's position

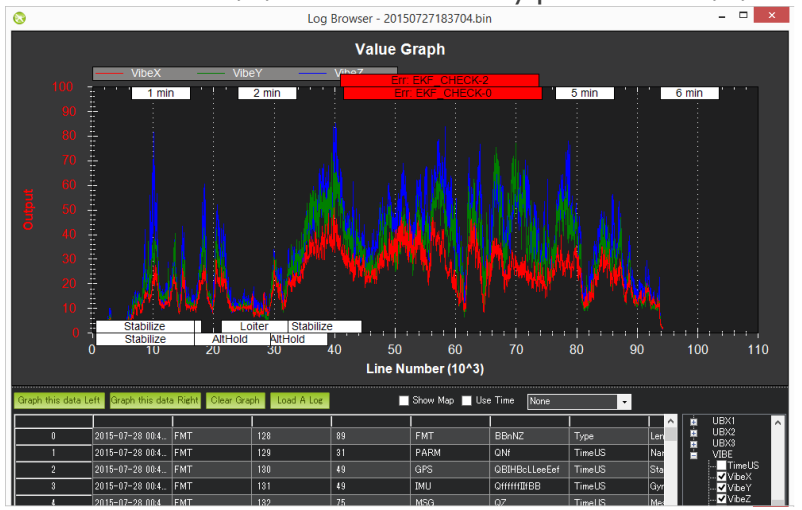






shows normal levels which are below 15m/s/s but occasionally peak to 30m/s/

VIBE[0] VibeZ (Min: 0.01 Max: 69.5 Mean: 5.53)
VIBE[0] VibeY (Min: 0.01 Max: 35.03 Mean: 2.26)
VIBE[0] VibeX (Min: 0.01 Max: 37.58 Mean: 1.59)
VIBE[1] VibeZ (Min: 0.06 Max: 73.37 Mean: 6.42)
VIBE[1] VibeY (Min: 0.04 Max: 42.83 Mean: 3.04)
VIBE[1] VibeX (Min: 0.04 Max: 44.63 Mean: 2.42)

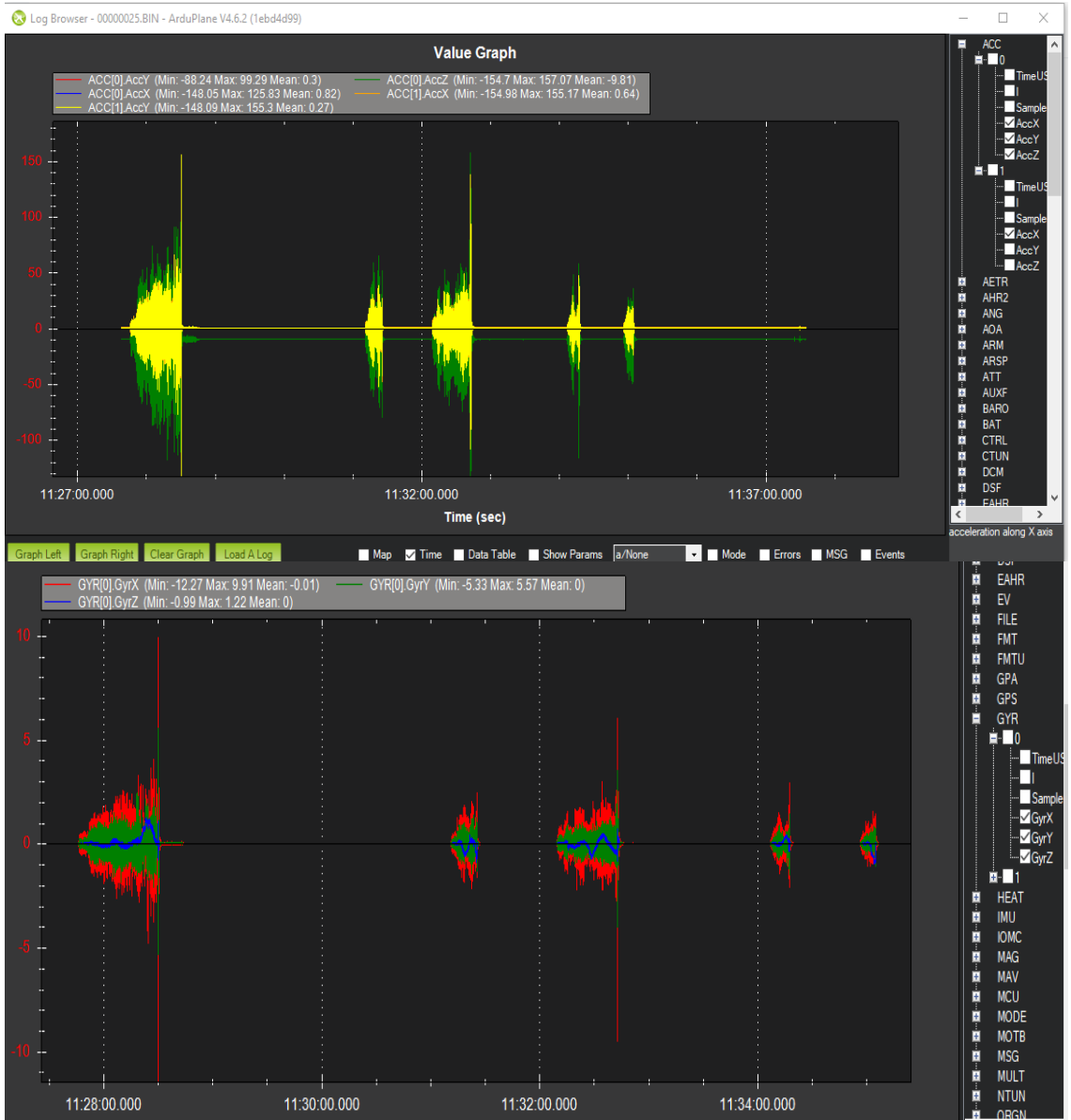



This is an example of a vehicle that had position estimate problems due to high vibrations.

- ✓ **Critical: VibeZ peaks 69.5 & 73.37** very high impulsive shocks.
- ✓ **Other axes:** VibeX/VibeY peaks up to ~44; means all axes see strong vibration, Z worst.
- ✓ **Interpretation:** not normal hover vibration, likely **prop/prop-mount issues, motor shaft/bearing play, or hard impacts/transmitted shocks to FC** (Z axis dominated). Gyro quietness from previous logs + high VIBE = impulsive shocks

# Measuring Vibration from Log

Check the scale on the left and ensure that your vibration levels for the **AccX and AccY are between -3 and +3**. For **AccZ the acceptable range is -15 to -5**.







- Check the scale on the left and ensure that your vibration levels for the AccX and AccY are between -3 and +3. For AccZ the acceptable range is -15 to -5. If it is very close or over these limits you should refer back to the Vibration Damping page for possible solutions.

ACC[0].AccY (Min: -88.24 Max: 99.29 Mean: 0.3)
ACC[0].AccX (Min:-148.05 Max: 125.83 Mean: 0.82)
ACC[1] AccY (Min:-148.09 Max: 155.3 Mean: 0.27)
ACC[0].AccZ (Min:-154.7 Max: 157.07 Mean: -9.81)
ACC[1] AccX (Min-154.98 Max: 155.17 Mean: 0.64).

**Interpretation :** these are **catastrophic shock magnitudes** on all axes; two accel sensors show similar large p–p → whole-airframe mechanical/prop/motor problem (not a single bad IMU).

GYR[0].GyrX (Min:-12.27 Max: 9.91 Mean: -0.01)
GYR[0] GyrZ (Min: -0.99 Max: 1.22 Mean: 0)
GYR[0] GyrY (Min: -5.33 Max: 5.57 Mean: 0) [1]

# Setup the “First Notch Filter”

**Goal : Filter out the lift-motor vibration peak (and its harmonics) so the attitude controller isn’t fighting noise; especially in Q-modes.**

**(mechanical fixes → logging → FFT → set notches → verify → tune)**

**Quick summary (what we’ll do)**

1.Fix the obvious mechanical vibration sources first (props, motors, mountings).

2.Collect high-rate IMU data (raw IMU or IMU batch sampling) while the aircraft hovers.

3.Use ArduPilot’s Filter/FFT tools to find the dominant gyro frequency and bandwidth.

4.Enable and configure the **Harmonic Notch** (INS_HNTCH_*) using the measured values.

5.Re-test, verify the FFT / Vibe graphs, then relax or raise gyro low-pass (INS_GYRO_FILTER) and finish PID tuning.

(ArduPilot docs with full details: see the Notch/QuadPlane tuning pages below.) Managing Gyro Noise with the Dynamic Harmonic Notch Filters — Plane documentation

# Setup the “First Notch Filter”

## Mechanical checklist (do these first)

These are the most common causes. Fixing them is faster and safer than trying to over-filter in software.



1. **Tighten** all prop nuts, motor screws and motor mounts.
2. **Balance props** (static balancing at minimum; dynamic if possible).
   Badly unbalanced big props = lots of aliased noise.

1. **Check motor bearings/shafts** for play or wobble. Replace suspect motors.

2. **Stiffen the frame / booms** (remove flexy arms, add braces). X-frames can twist under thrust check for twist under load.



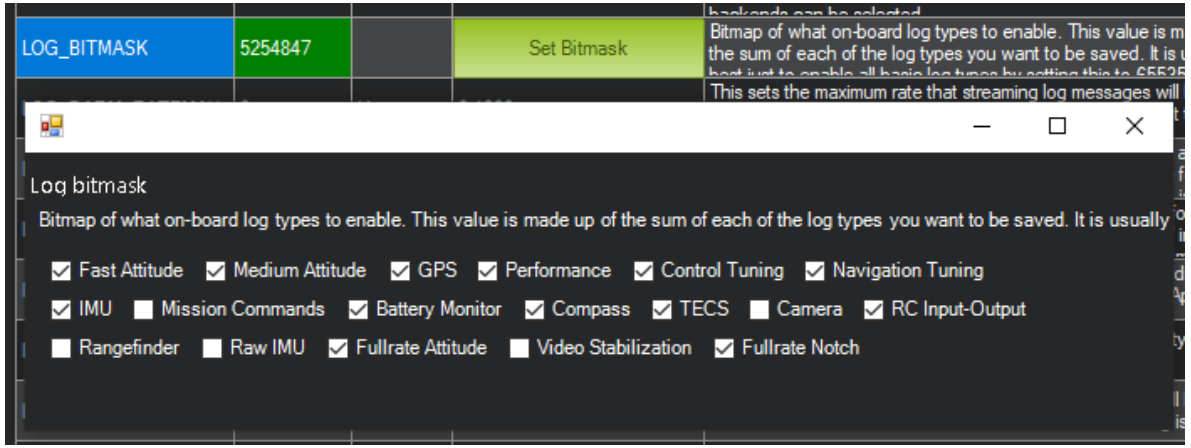
3. **Flight controller mounting**: use a recommended mount or short soft standoffs, but not an overly floppy mount that introduces resonance. If you move the FC, update IMU position params **(INS_POS*).**

4. **Route power wires** and heavy wiring away from the FC/IMU and GPS.
   (If vibration remains high after mechanical work, proceed to logging + notch filters.)
   ArduPilot.org

# Setup the “First Notch Filter”

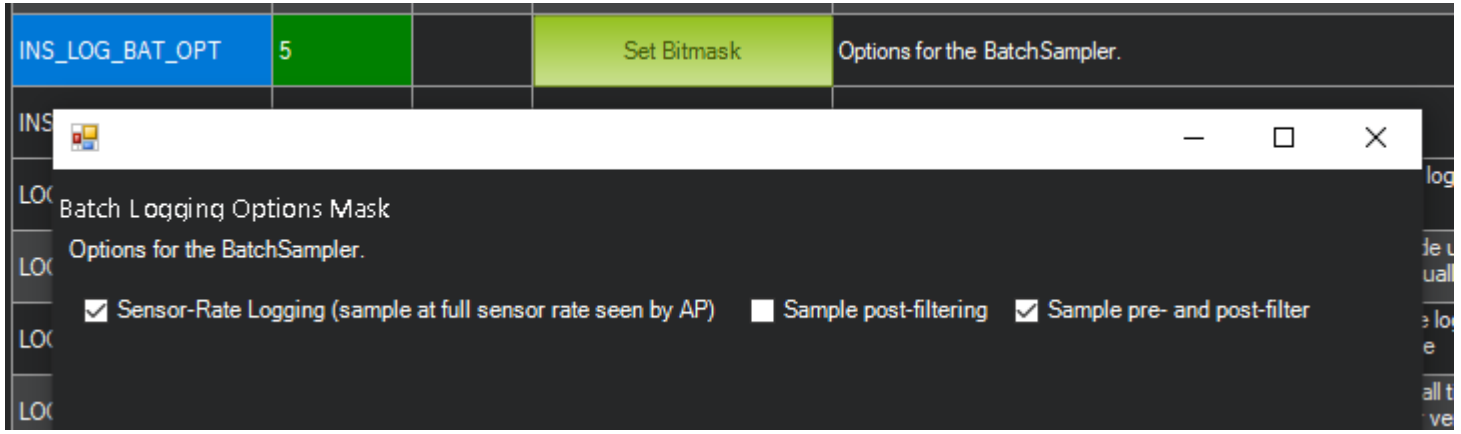
## 1: Setup - Logging

FFT Setup page -> **Sample count = 2048**




**LOG_BITMASK = 5254847**

We must capture pre-filter IMU data so you can identify the motor/prop frequency.




**INS_LOG_BAT_OPT 5**

**INS_LOG_BAT_MASK = 1** (log IMU1)

# What we can do

## 1: Setup – Harmonic Notches FFT

**Setup the “First Notch Filter”**

**Goal : Filter out the lift-motor vibration peak (and its harmonics) so the attitude controller isn’t fighting noise; especially in Q-modes.**

**INS_HNTCH_ENABLE = 1** # turn on harmonic notch → reboot.
**INS_HNTCH_MODE = 4** # 1=Throttle tracking (works without RPM/ESC data)
**INS_HNTCH_HMNCS = 7** # fundamental + 2 harmonics
**INS_HNTCH_ATT = 40** # attenuation (30–50 typical)
**INS_HNTCH_REF = 1** no scaling required

# Initial frequency guess for heavy lift quads (18–22" props typically hover ~60–100 Hz)
**INS_HNTCH_FREQ = FFT_MINHZ = 20 Hz**
**INS_HNTCH_BW = INS_HNTCH_FREQ /2 = 10hz**
**INS_HNTCH_FM_RAT = 1** prevent the notch going below INS_HNTCH_FREQ
# Logging (helps you verify & fine-tune)
**LOG_BITMASK** = 131071 # or ensure IMU raw/fast is enabled in your GCS

**INS_GYRO_FILTER = 20** # Safe for 25" props
**INS_ACCEL_FILTER = 10**
**Q_A_RAT_RLL_FLTD = 10**
**Q_A_RAT_PIT_FLTD = 10**

## Static notch for frame resonance

**INS_NOTCH_ENABLE = 1**
**INS_NOTCH_FREQ1 = <frame frequency from FFT, e.g. 40>**
**INS_NOTCH_BW1 = 8–12 (narrow)**
**INS_NOTCH_ATT1 = 18–25**
**Write params and reboot.**

# What we can do

## 1: Setup – Harmonic Notches - Throttle

**Setup the “First Notch Filter”**

**Goal : Filter out the lift-motor vibration peak (and its harmonics) so the attitude controller isn’t fighting noise; especially in Q-modes.**

**INS_HNTCH_ENABLE = 1** # turn on harmonic notch
**INS_HNTCH_MODE = 1** # 1=Throttle tracking (works without RPM/ESC data)
**INS_HNTCH_HMNCS = 4** # fundamental + 2 harmonics
**INS_HNTCH_ATT = 40** # attenuation (30–50 typical)
**INS_HNTCH_REF = Q_M_THST_HOVER** # set equal to your hover throttle (e.g. 0.30)

**INS_HNTCH_FREQ = 80 # Starting center frequency (Hz)**
**INS_HNTCH_BW = INS_HNTCH_FREQ /2 = 40hz** # Bandwidth (half of freq)

**INS_HNTCH_FM_RAT = 0.7** # Tracks down to ~70% of hover
**INS_HNTCH_OPTS = 0** # Leave advanced options off
# Logging (helps you verify & fine-tune)
**LOG_BITMASK** = 131071 # or ensure IMU raw/fast is enabled in your GCS

if Throttle based make sure your motor voltage compensation is setup (see Q_M_BATT_x params for QuadPlane)

# (Throttle-Based Harmonic Notch) vs (In-Flight FFT)

**INS_HNTCH_MODE = 1 (Throttle-Based Harmonic Notch):**

- **How it works:** This mode scales the notch filter's center frequency based on the main throttle output. It assumes a relationship between throttle percentage and motor RPM/noise frequency.

•Pros: Easy to set up, no additional hardware required.
•Cons: Less accurate than RPM-based or FFT-based methods as the relationship between throttle and motor frequency isn't perfectly linear or constant across different flight conditions. It's an estimation.

**INS_HNTCH_MODE = 4** (In-Flight FFT Based Harmonic Notch):

- **How it works:** The autopilot performs a Fast Fourier Transform (FFT) on the raw gyro data in real-time to identify the dominant noise frequencies and then places the notch filters dynamically based on these findings.

•Pros: Highly accurate as it directly measures noise frequencies. Works well on H7 boards like your Pixhawk 6C which have the processing power.
•Cons: More CPU intensive than throttle-based. Requires FFT_ENABLE = 1 and setting up FFT_MINHZ, FFT_MAXHZ, and FFT_WINDOW_SIZE correctly.

**Recommendation for Pixhawk 6C (H7):**
**For the Pixhawk 6C with an H7 processor, the recommendation is INS_HNTCH_MODE = 4 (In-Flight FFT). This method provides better performance and more accurate noise suppression compared to the throttle-based method, because of the flight controller's power.**

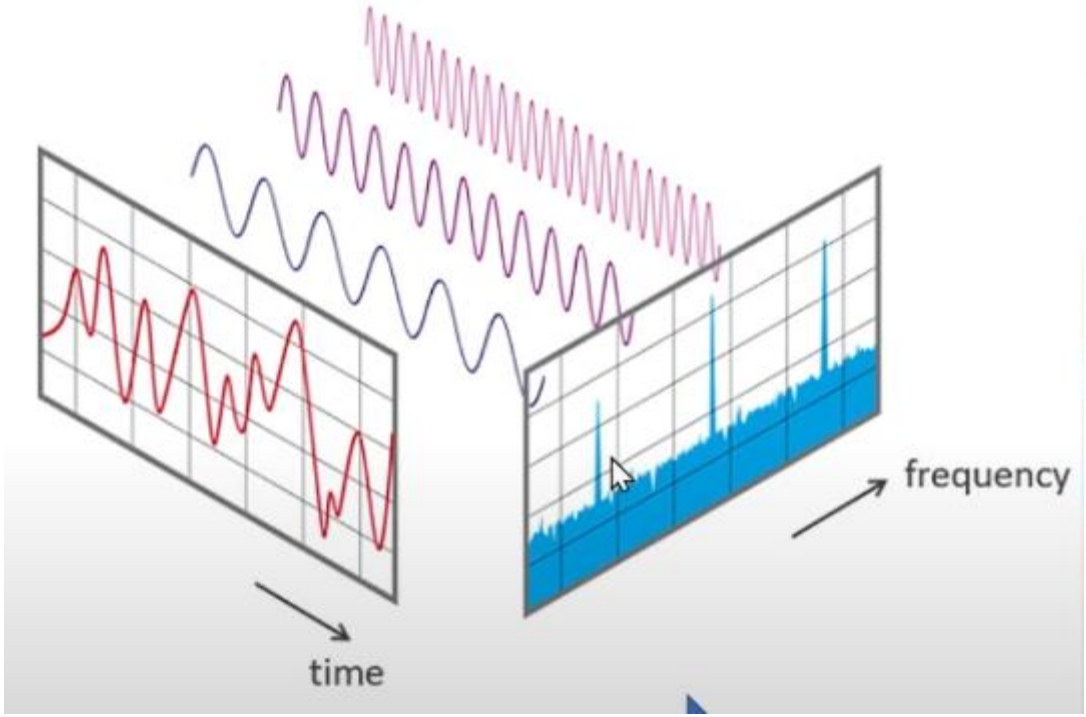
# Setup the “First Notch Filter”

## 1: Setup – Dynamic FFT (Fast Fourier Transform)

Set **FFT_ENABLE = 1** to enable FFT and show parameters
Set **INS_HNTCH_MODE = 4** to use FFT to drive the notches
**INS_HNTCH_REF:** Set to 1
Set **FFT_MAXHZ** = X
X = 0.8*Vbatt*KV/60
**Vbatt=50.4 V**
**KV=180 (from T-Motor AT8020 KV180)**

X = 0.8*50.4*180/60
Y = 500 / 25

Set **FFT_MINHZ** = Y
Y = 500 / Dprop
**Dprop = Prop Diameter in Inches = 25**
**FFT_MAXHZ = 120.96 Hz**
**FFT_MINHZ = 20 Hz**
**FFT_NUM_FRAMES = 0** FFT needs to be reactive to changing RPM
**FFT_OPTIONS = 1** to apply low pass filters before doing the FFT
**FFT_WINDOW = 1024** if CPU load allows or as high as possible.



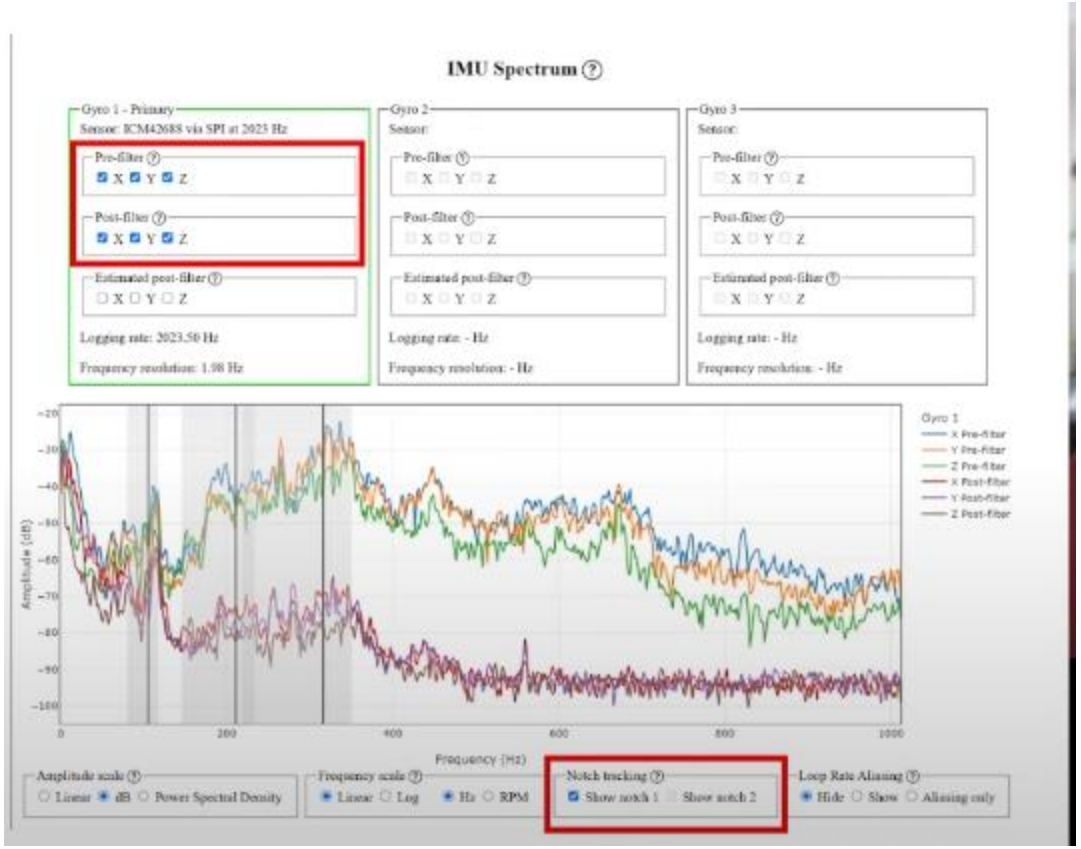

Setting FFT_WINDOW_SIZE for Pixhawk 6C (H7 Processor),
you should set **FFT_WINDOW_SIZE = 512** (If your Pixhawk 6C's CPU speed allows, you will benefit from FFT_WINDOW_SIZE = 1024)

# Checking – Filter Review tool

## 1: Setup – IMU Spectrum

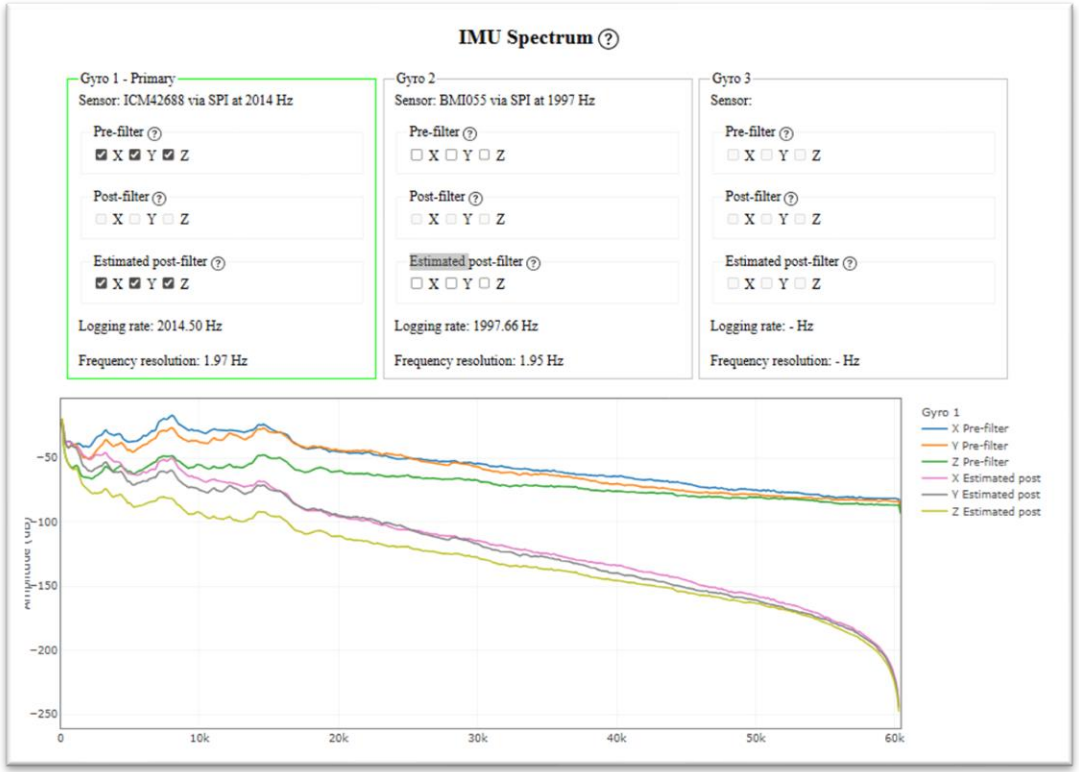
With the logging settings we should be able to see

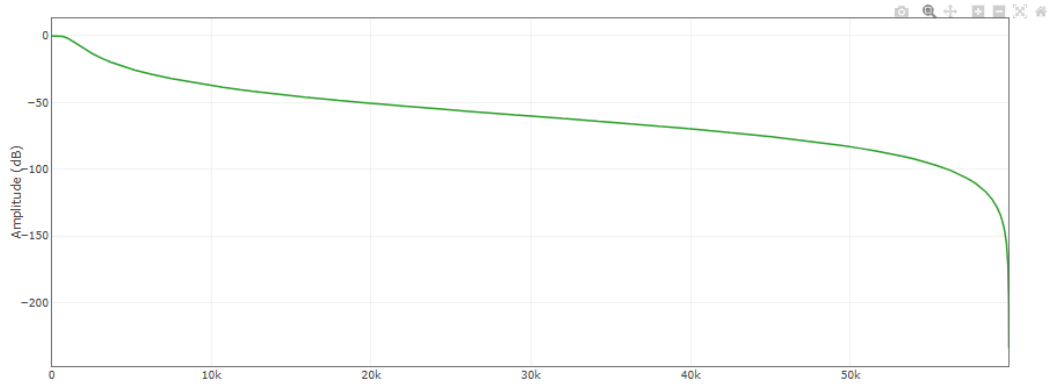
- Pre-Filter Gyro Data
- Post-Filter Gyro Data
- Notch Tracking



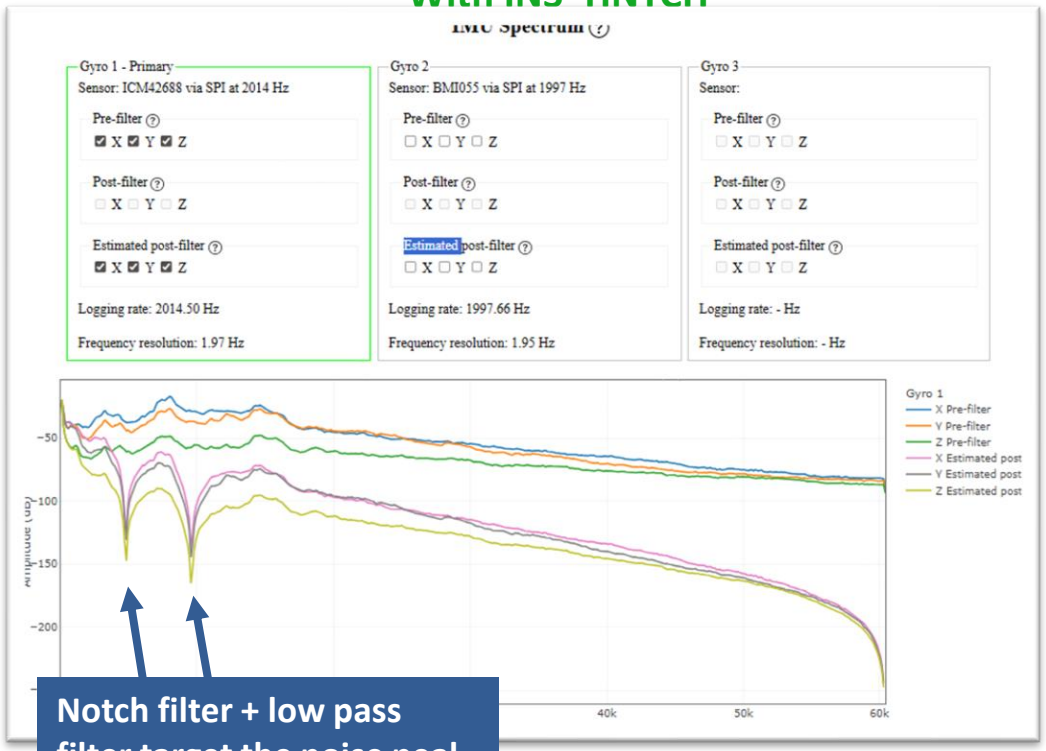
# To understand Harmonic Notch Filter better

## Without INS_HNTCH_

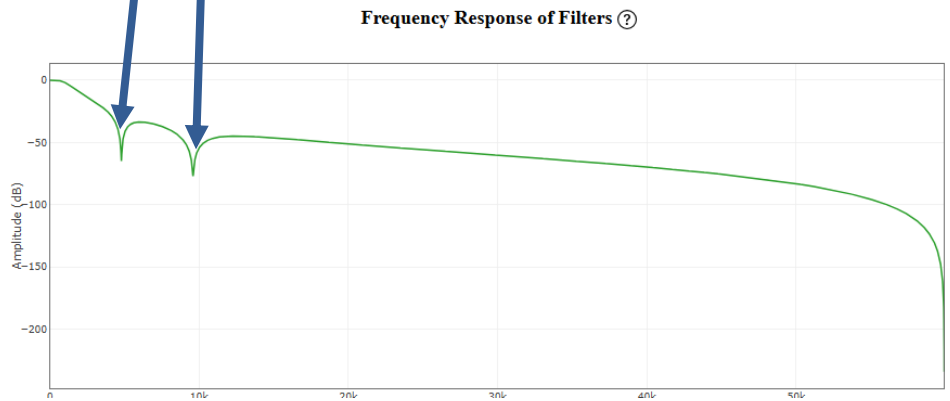



## With INS HNTCH




**Notch filter + low pass filter target the noise peal**



# Questions for Mr.Marc

T-Motor AT8020

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| AT8020 KV180 | 25*10 | 40% | 44.68 | 11.61 | 518.70 | 2970 | 1.140 | 3277 | 6.30 | 93℃ |
| | | 45% | 44.31 | 15.40 | 682.40 | 3330 | 1.440 | 4226 | 6.20 | |
| | | 50% | 44.35 | 19.36 | 858.60 | 3600 | 1.720 | 5154 | 6.00 | |
| | | 55% | 44.16 | 24.25 | 1070.90 | 3870 | 2.010 | 6120 | 5.70 | |
| | | 60% | 44.07 | 30.09 | 1326.10 | 4170 | 2.370 | 7307 | 5.50 | |
| | | 65% | 43.72 | 37.41 | 1635.60 | 4500 | 2.810 | 8699 | 5.30 | |
| | | 70% | 43.77 | 45.69 | 1999.90 | 4830 | 3.230 | 10085 | 5.00 | |
| | | 75% | 43.26 | 53.95 | 2333.90 | 5100 | 3.620 | 11455 | 4.90 | |
| | | 80% | 43.17 | 63.48 | 2740.40 | 5400 | 4.030 | 12741 | 4.60 | |
| | | 90% | 42.40 | 84.18 | 3569.20 | 5880 | 4.890 | 15371 | 4.30 | |
| | | 100% | 41.53 | 111.25 | 4620.20 | 6330 | 5.800 | 18121 | 3.90 | |





# Test flight & verification

1. Hover in a calm area (QHOVER / QSTABILIZE) for ~30–120 s with minimal stick input.
2. Download .bin log. Open with Filter Review or Mission Planner's FFT tools and compare pre- and post-filter spectra, the main gyro peak should be greatly reduced. Also check **the Vibe graph in Mission Planner (VibeX, VibeY, VibeZ). Vibe < ~30 m/s/s is normally acceptable; >60 m/s/s usually causes problems**. ArduPilot.org+1
3. Watch CPU load: **PM.Load and PM.NLon** in the log. If CPU load is very high after enabling notches, **reduce the number of notches/harmonics** or disable per-IMU notches. The docs recommend staying under ~60% CPU load for heavy FFT use.